Zachary Mineiko

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Education

University of Iowa, College of Engineering — May 2025
B.S. in Electrical Engineering, Computer Science Focus — *Iowa City, IA*
Coursework: Embedded Systems, Linear Systems, Computer Architecture, Comm. Networks, Generative Artificial Intelligence

Technical Skills

Circuit Design & Simulation: ModelSim, KiCad, Autodesk Fusion 360, Micro-Cap SPICE, Atmel Studio
Hardware & Embedded: ESP32, AVR Assembly, Arduino, Platform IO, RFID, PCB Design, Analog/Digital Signal Processing
Languages & Tools: Python, C/C++, Embedded C, Verilog, Bash, MATLAB, OpenCV, Matplotlib, SolidWorks

Experience

Full Stack Engineer (Contract) — Aug 2025 – Present
FlipWyz — *Remote*

- Architect and design a production ready cross platform marketplace web app with React Native and Supabase
- Implement modular and efficient component abstractions to reduce render overhead and improve responsiveness
- Apply analytical and data driven problem solving to enhance query performance and reduce latency

Engineering Technician Intern — Mar 2022 – Aug 2022
Firefly — *Chicago, IL*

- Diagnosed and resolved firmware and hardware failures in deployed digital billboard systems, reducing defect recurrence through systematic root cause analysis and corrective action
- Transformed Chicago branch from lowest to highest performing location by implementing structured diagnostic and repair methodologies across the hardware fleet
- Engineered data collection and process automation workflows using Python, OpenCV, and Matplotlib, increasing daily throughput of repaired units by 20%
- Developed and maintained standardized repair procedures to improve test yield consistency and reduce repeat failures
- Participated in Agile development processes through weekly stand-ups, contributing to cross-functional team coordination

Electrician's Assistant / Web Developer — May 2020 – May 2023
Custom Craft Builders — *Chicago, IL*

- Led installation teams for low-voltage electrical cable systems including coaxial and Ethernet network infrastructure
- Interpreted electrical diagrams and wiring specifications to ensure accurate, code compliant installations
- Coordinated with project leads to schedule and execute multi-site installations efficiently

Projects

Smart Pet Feeder – Senior Design Project | *ESP32, Embedded C, React, Autodesk Fusion 360, PostgreSQL*

- Developed a multi-voltage ESP32 based IoT control system with real-time sensor data processing, wireless connectivity, and individualized feeding schedules for multi-pet households
- Created custom PCB schematics and component footprints using Autodesk Fusion 360, integrating multiple sensors into a unified hardware platform
- Designed, 3D modeled, and printed the majority of the physical enclosure and mechanical assemblies, ensuring dimensional tolerances for proper component fit, sensor alignment, and repeatable full system manufacturability
- Designed and validated system architecture end-to-end from hardware test bench through embedded firmware to data backend

IR Receiver with Real-Time Signal Processing | *ESP32, Embedded C, Analog & Digital Filtering*

- Engineered a hybrid analog/digital IR beam interruption detection system using a 940nm photodiode and ESP32 ADC sampling for real-time signal acquisition
- Designed multi-stage analog signal conditioning with non-inverting amplification, buffering, and a 2nd-order bandpass filter centered at 421Hz to match emitter frequency constraints
- Implemented a 6th order Chebyshev digital filter to extract signal envelope and reject ambient noise, achieving reliable detection up to 42 inches in high noise environments
- Validated system performance across environmental conditions, confirming detection stability and low-latency response

SISC Computer – 32-bit RISC Processor | *Verilog, ModelSim, HDL*

- Designed and implemented a complete 32-bit RISC-style processor in Verilog with custom datapath, FSM control unit, and integrated memory subsystems
- Validated processor functionality through ModelSim simulation, executing test algorithms including bubble sort